

Article Sections: Abstract, Discussion, and Conclusion

Abstract

This study extends prior work on deictic framing and large language model (LLM) moral reasoning by comparing a published English baseline with a full unrestricted Yoruba corpus across the same 6 dilemmas $\times$ 9 deictic framings $\times$ 3 model families design. The central question is whether deictic framing exerts comparable effects on ethical response in a typologically distinct language, and whether any observed differences are better explained by preferred-solution shifts, rhetorical shifts, or language-instability effects. To address this, we generated a full open-condition Yoruba corpus and paired it cell-by-cell with the published English baseline by (model, dilemma_id, framing_type). We then coded the Yoruba responses directly from the raw Yoruba text using a multi-layer scheme that captured preferred solution, ethical preference type, response genre, deictic uptake quality, and language stability, and aligned this with the repository's original English-side coding dimensions.

The results show that unrestricted Yoruba is fully comparable to English at the level of experimental design, but that the strongest cross-linguistic differences are not reducible to final ethical choice alone. GPT-4o in open Yoruba remained largely framework-expository and conditionally committed, suggesting broad ethical adjacency to English while shifting toward more procedural and advisory discourse. Claude was the most action-codable model in Yoruba, showing the widest spread across utilitarian, deontological, procedural-caution, care-ethics, and mixed responses, and often becoming more directive than its English counterpart. DeepSeek was the most unstable, with frequent translation-mode or meta-commentarial drift that complicated stance coding and increased adjudication burden. Severe language problems were minimal for GPT-4o (1/54) and Claude (2/54) but substantial for DeepSeek (20/54). We argue that open Yoruba and English should be compared not only in terms of preferred solution, but across a multi-dimensional discourse space including ethical framing, rhetorical posture, deictic uptake, and response stability. On this basis, the open Yoruba corpus provides a robust and publishable cross-linguistic extension of the original English study.

Discussion

The most important finding of the unrestricted Yoruba condition is that deictic framing continues to reorganize model output, but not always by shifting the final ethical side alone. In many cases, the more consequential shift is in the **mode of moral response**. The English baseline, as originally analyzed in the repository, is predominantly characterized by mixed ethical framing, analytical affective stance, and a moral-analyst or guide-like voice that often withholds a single final commitment. Open Yoruba frequently departs from this pattern by moving into procedural advice, direct recommendation, or unstable role-exit behavior, depending on the model. The consequence is that cross-linguistic comparison cannot be reduced to a binary question such as “did English and Yoruba choose the same option?” Rather, the relevant empirical space is multi-layered: deictic framing may alter preferred solution, ethical architecture, rhetorical authority, or response genre, and these dimensions do not always move together.

GPT-4o illustrates the most conservative form of cross-linguistic divergence. In unrestricted Yoruba, it remained overwhelmingly coded as `balanced_framework_exposition` with `conditional_or_mixed` preferred solutions. This suggests that, at the broad moral level, GPT-4o frequently occupies the same ethical neighborhood in both languages: it still weighs competing considerations, still avoids overcommitting in many cells, and still treats complex dilemmas as sites for structured moral reflection. Yet it does not do so in exactly the same rhetorical manner. Relative to English, open Yoruba GPT-4o often sounds more procedural, more role-bound, and more oriented toward practical next steps. This means that the model is not simply translating its English reasoning habits into Yoruba. Rather, the Yoruba condition appears to tilt the model toward a more actionable advisory mode even when it does not force a clean verdict.

Claude shows the strongest evidence that deictic framing in Yoruba can shift not just discourse form, but **ethical explicitness**. Among the three models, Claude was the most substantively classifiable in the unrestricted Yoruba corpus, with meaningful distributions across `supports_A`, `supports_B`, `conditional_or_mixed`, and `refuses_to_commit`, and with the richest spread of ethical preference types. This is especially important because the corresponding English baseline for Claude often remains restrained, neutral, or analytically noncommittal. The open Yoruba results therefore suggest that certain Yoruba framings—especially those with clearer role anchoring or practical address—can encourage Claude to cross the threshold from reflection into recommendation. That does not imply that Yoruba simply “makes Claude more decisive.” Rather, it suggests that the interaction of Yoruba

deictic marking and moral perspective-taking can lower the rhetorical barrier to explicit normative action.

DeepSeek is the strongest cautionary case. It is not simply “worse” than the other models; it is different in a way that is theoretically relevant. In a substantial minority of open Yoruba cells, DeepSeek leaves the expected response frame and shifts into translation, glossing, meta-commentary, or fragmented analysis. This means that some of its largest apparent cross-linguistic divergences are not best understood as differences in ethical preference, but as failures of stable enunciative uptake. In other words, the most revealing DeepSeek result is often not “it chose a different ethical side than English,” but “it did not remain inside the same kind of moral speech act that the English baseline maintained.” For cross-linguistic AI ethics research, this is a meaningful finding: the capacity to inhabit a framed moral position is itself part of the phenomenon under study.

These results also clarify the role of language stability in the comparative interpretation. The open Yoruba corpus is not uniformly compromised by mixed-language drift. GPT-4o and Claude remain highly usable for substantive comparison, with only 1/54 and 2/54 severe language-problem cells respectively. This means that most of the differences seen for those models are best interpreted as genuine cross-linguistic or cross-rhetorical effects rather than as simple noise. DeepSeek, by contrast, requires a bifurcated interpretation: some differences are substantive, while others are consequences of response-mode instability. This is precisely why the unrestricted Yoruba corpus needed a raw-Yoruba coding layer with adjudication support. Without that step, one would risk conflating true ethical divergence with unstable language behavior.

Methodologically, the study also demonstrates that the repository’s original English coding scheme and the new raw-Yoruba coding scheme can be aligned, but only if one treats both as discourse-analytic rather than verdict-only systems. The repository’s English side already encoded dimensions such as `primary_framework`, `ethical_reasoning_type`, `voice_authority`, `moral_reasoning`, `affective_stance`, and `indexical_coherence`. The Yoruba coding introduced aligned but not identical dimensions: `ethical_preference_type`, `response_genre`, `deictic_uptake_quality`, and `language_stability`, alongside `preferred_solution`. The strongest comparison emerges when these layers are read together. In this sense, the unrestricted Yoruba corpus does not merely add another language to the project; it forces the analysis to become more explicit about what exactly counts as ethical similarity across languages.

Finally, these findings bear directly on the article’s original theoretical question. If deictic framing reorganizes LLM moral reasoning, then a cross-linguistic test should reveal not only whether different languages yield different moral recommendations, but whether they

shape the very *form* of enunciation differently. The open Yoruba corpus supports exactly that claim. The most informative differences between English and Yoruba are often not binary conflicts of answer, but shifts in whether the model speaks as analyst, guide, institutional advisor, procedural actor, translator, or unstable commentator. That is not a methodological nuisance. It is evidence that deictic framing and language structure jointly mediate how moral reasoning becomes text.

Conclusion

The unrestricted Yoruba corpus confirms that English and Yoruba can be compared meaningfully within the same deictic ethical design, but that the comparison must be carried out on more than one level. At the level of broad ethical orientation, the two languages are often adjacent: both can remain mixed, both can enumerate competing principles, and both can withhold definitive commitment. At the level of rhetorical realization, however, the differences are substantial. Open Yoruba more often becomes advisory, procedural, or directive, while English more often remains analytically expository. These differences are strongest and most interpretable for GPT-4o and Claude, and most unstable for DeepSeek.

The study therefore supports a multi-dimensional account of cross-linguistic moral response in LLMs. Deictic framing does not simply determine which ethical choice is made; it also shapes how moral subjectivity is staged, how responsibility is voiced, and whether the model responds as a decision-maker, a theorist, or an unstable mediator. This makes unrestricted Yoruba not only a valid comparison corpus, but a theoretically revealing one. The corpus shows that cross-linguistic AI ethics research must track preferred solution, ethical preference type, response genre, deictic uptake, and language stability together. When analyzed in that way, the English-Yoruba comparison yields a clear and insightful result: the deictic organization of ethical response is both structurally persistent across languages and materially transformed by the linguistic resources through which it is expressed.